

Vol 8 Chapter 8 — Oscillations and Normal Modes

Keeper (author pass — deep math/physics revision)

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Chapter 8 — Oscillations and Normal Modes

Level 1 — one sentence

Oscillations — simple, damped, driven, coupled — appear everywhere in physics because near any stable equilibrium the potential is well-approximated by a parabola, giving simple harmonic motion locally, and the eigenvalue analysis of multi-DOF systems yields normal modes that decouple the dynamics into independent oscillators.

Level 2 — graduate-physicist precision

8.1 Simple harmonic motion (SHM)

A particle in potential $V(x) = (1/2)kx^2$ has equation of motion $m\ddot{x} = -kx$, giving:

$$x(t) = A \cos(\omega t + \phi), \quad \omega = \sqrt{k/m}$$

Constants A (amplitude) and ϕ (phase) set by initial conditions.

Energy: $E = (1/2)kA^2$. Always conserved (no dissipation).

8.2 Damped oscillator

With viscous damping $-b\dot{x}$:

$$m\ddot{x} + b\dot{x} + kx = 0$$

Solutions classified by $\zeta = b/(2\sqrt{mk})$: - Underdamped ($\zeta < 1$): $x = Ae^{-\zeta\omega_0 t} \cos(\omega_d t + \phi)$ with $\omega_d = \omega_0\sqrt{1 - \zeta^2}$ — oscillating decay - Critically damped ($\zeta = 1$): fastest return to equilibrium without oscillation - Overdamped ($\zeta > 1$): two real exponential decays, slow return

Quality factor $Q = \omega_0 m/b = 1/(2\zeta)$.

8.3 Driven oscillator and resonance

With external drive $F_0 \cos(\omega t)$:

$$m\ddot{x} + b\dot{x} + kx = F_0 \cos(\omega t)$$

Steady-state response:

$$x(t) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (b\omega/m)^2}} \cos(\omega t - \delta)$$

with phase lag δ . Resonance at $\omega = \omega_d \approx \omega_0$; amplitude diverges for zero damping. The Q-factor sets the sharpness ($\text{FWHM} = \omega_0/Q$).

8.4 Normal modes

For a system of N coupled oscillators: equations $M\ddot{\vec{x}} = -K\vec{x}$ with mass matrix M and stiffness matrix K .

Try $\vec{x} = \vec{a}e^{i\omega t}$: eigenvalue problem $(K - \omega^2 M)\vec{a} = 0$. Eigenvalues ω_i^2 are squared normal-mode frequencies; eigenvectors $\vec{a}_i$ are normal mode shapes.

Each normal mode oscillates independently at its own frequency. General motion is superposition of normal modes.

8.5 Worked example: coupled pendulum

Two identical pendulums of length ℓ coupled by a spring with constant k :

$$\begin{aligned} m\ell^2\ddot{\theta}_1 &= -mg\ell\theta_1 - k\ell^2(\theta_1 - \theta_2) \\ m\ell^2\ddot{\theta}_2 &= -mg\ell\theta_2 - k\ell^2(\theta_2 - \theta_1) \end{aligned}$$

Normal modes: - Symmetric: $\theta_1 = \theta_2$, frequency $\omega_+ = \sqrt{g/\ell}$ (gravity alone, spring uncoupled) - Antisymmetric: $\theta_1 = -\theta_2$, frequency $\omega_- = \sqrt{g/\ell + 2k/m}$ (gravity + spring)

If you displace one pendulum and release: motion alternates between the two — energy transfers back and forth at the beat frequency $|\omega_+ - \omega_-|$.

8.6 Continuum oscillations

A string of length L with linear density μ under tension T : wave equation

$$\partial_t^2 y = (T/\mu)\partial_x^2 y$$

with wave speed $v = \sqrt{T/\mu}$.

Standing-wave normal modes: $y_n = A_n \sin(n\pi x/L) \cos(\omega_n t)$ with $\omega_n = n\pi v/L$. Harmonic series (Pythagoras).

8.7 Substrate connection

Each normal mode is the substrate's natural K-type at that frequency. The substrate's discrete K-type structure naturally produces discrete normal-mode spectra; continuous-medium normal modes are the substrate K-type spectrum in the appropriate boundary-condition sector.

8.8 K-audit anchors

- Vol 5 Ch 4: quantum harmonic oscillator (substrate K-type analog)
- Vol 6 Ch 7: phonons (quantized normal modes of solids)
- Vol 9 Ch 7: phonons in condensed matter

Level 3 — 5th-grader accessibility

Anything that has a stable equilibrium and gets nudged slightly oscillates — pendulums, mass-on-spring, atoms in molecules, the wing of a plane in turbulence. Near equilibrium, $V \approx (1/2)kx^2 \rightarrow$ simple harmonic motion with frequency $\omega = \sqrt{k/m}$. Add damping $\rightarrow$ oscillation decays (underdamped) or just settles (critically/overdamped). Add driving force at frequency $\omega \rightarrow$ resonance peak when $\omega = \omega_0$. Multiple coupled oscillators decompose

into independent normal modes at characteristic frequencies. A vibrating guitar string has a harmonic series of normal modes — that's the music. In BST, every normal mode is a substrate K-type at its natural frequency.

What comes next

Chapter 9 develops continuum elasticity.

Where to look this up

- Goldstein Ch 6; Marion-Thornton Ch 11-12
- BST: Vol 5 Ch 4 (quantum HO); Vol 9 Ch 7 (phonons)